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Machine Learning for Genotype × Environment Interactions in Nigerian Maize Breeding Programs

Practical Machine Learning Final Project
Master's in Green Data Science

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1. Introduction

Maize (*Zea mays* L.) is the most important cereal crop in Nigeria, serving as a staple food for millions of households and a critical raw material for the poultry and livestock feed industries. A major challenge in the development and commercialisation of improved varieties is genotype × environment (G×E) interaction, whereby the relative performance of genotypes changes across environments — a genotype that yields best in one location may not maintain that ranking elsewhere. To address this, breeders conduct multi-environment trials (METs), evaluating candidate genotypes across multiple locations, management conditions, and years before variety release decisions can be made. These trials are scientifically necessary but expensive, labour-intensive, and time-consuming, often requiring three to five years of field evaluation across numerous testing environments.

Machine learning (ML) offers an opportunity to complement conventional breeding approaches by modelling the complex, non-linear relationships among genotypes, environments, and management practices that classical linear methods may not fully capture. Using historical breeding data, ML models can help predict genotype performance in untested environments and surface patterns not readily visible through traditional statistical analysis.

This project applies ML to model G×E interactions in Nigerian maize breeding trials, providing a decision-support framework to help breeders reduce the cost and duration of field evaluation while maintaining the rigour required for variety release. The specific objectives were to: (1) train and evaluate supervised ML models for predicting maize grain yield in untested environment–management combinations, and (2) apply unsupervised learning to identify trial locations that provide redundant environmental information. The best-performing model was deployed as an interactive Streamlit application to support breeder and researcher decision-making

2. Data

2.1 Dataset Description

The historical dataset used in this project was obtained from multi-environment maize breeding trials conducted in Nigeria between 2020 and 2022 by four breeding institutions: Institute for Agricultural Research (IAR), University of Ilorin, International Maize and Wheat Improvement Center (CIMMYT), and International Institute of Tropical Agriculture (IITA). The raw dataset comprised 21,330 observations representing 237 maize genotypes evaluated across five locations spanning three agroecological zones, under three management conditions (Appendix A).

The dataset contains agronomic, environmental, and management-related variables; some categorical, others continuous numerical or ordinal scores (Appendix B).

2.2 Data Cleaning and Preprocessing

The following preprocessing steps were performed before model development:

- Location name standardisation by stripping leading and trailing whitespace and converting names to title case.
- Environment identifier creation by concatenating location, management condition, and year.
- Missing value imputation using the median of the respective variable.
- Replication aggregation, where multiple plot-level replication records were averaged to generate a single observation for each genotype–environment combination.

2.3 Feature Selection

Feature selection was guided by a combination of domain knowledge and correlation analysis. Variables were selected based on their biological relevance to maize performance and GxE interactions. The final feature set comprised 33 variables, including six environmental covariates, four soil properties, eleven phenotypic traits, seven encoded categorical variables, and five engineered features (Appendix B).

2.4 Feature Engineering

Additional features were engineered to help the model capture G×E interaction structure:

- Genotype mean yield (geno_mean_yield): the average yield of each genotype across all environments, capturing the main genotype effect.
- Environment mean yield (env_mean_yield): the average yield within each environment, capturing the main environment effect.
- Location mean yield (loc_mean_yield): the average yield at each location across all conditions and years.
- Stress Tolerance Index (STI): computed as (yield_stress × yield_optimum) / $(\text{mean\_optimum\_yield})^2$, a standard plant breeding index that quantifies a genotype's productivity under stress relative to potential under optimum conditions.
- Yield coefficient of variation (geno_yield_cv): the standard deviation of a genotype's yield divided by its mean, expressed as a percentage. Lower CV indicates greater yield stability across environments.
- Categorical encodings: all categorical variables (agroecological zone, stress condition, season, maturity group, breeding institution, genotype name, and region) were encoded as integers using ordinal or label encoding.

3. Data Organization

To ensure robust model evaluation under different real-world breeding scenarios, the dataset was organised using three complementary cross-validation (CV) schemes rather than a single static train–test split.

a. **CV1 — Random 5-Fold Cross-Validation.** The dataset was shuffled and divided into five equally sized folds, with four folds used for training and one for validation in each iteration (K-Fold with shuffling). This evaluates overall predictive accuracy under the assumption that both genotypes and environments in the validation set were likely observed elsewhere during training.

b. **CV2 — Leave-One-Environment-Out.** A group-based cross-validation strategy was applied using the environment_id variable: in each iteration, all observations from a single environment (location × stress condition × year) were held out for validation, with the remaining 44 environments used for training. This simulates predicting performance at a new trial site or under a condition not previously encountered.

c. **CV0 — Leave-One-Genotype-Group-Out**. A second group-based strategy was implemented using genotype identity (Name): all observations belonging to a subset of genotypes were excluded from training and used exclusively for validation. This evaluates the model's ability to generalise to entirely new, unseen maize varieties, which is critical for plant breeding applications where prediction of novel genotypes is required.

4. Methods- Machine Learning Models used

4.1 Supervised Learning — Yield Prediction

Six models were compared: a classical plant breeding baseline and five machine learning models, selected to represent a progression from linear to non-linear and from interpretable to high-capacity.

- AMMI-2 (baseline): the standard model used in plant breeding for analysing G×E interactions via additive main effects and a rank-2 SVD interaction term, providing a benchmark against which ML methods are evaluated.
- Ridge Regression: a regularised linear model handling multicollinearity among agronomic and environmental predictors while remaining interpretable.
- Lasso Regression: L1 regularisation performs automatic feature selection, identifying the most influential predictors of grain yield.

- Support Vector Regression (SVR): models non-linear relationships through kernel-based learning.
- Random Forest: captures complex non-linear interactions, is robust to overfitting, and provides feature importance estimates.
- XGBoost: strong predictive performance on structured tabular data, modelling high-order interactions through gradient boosting.

Hyperparameter settings for all models are provided in Appendix C. All models except AMMI-2 were implemented within a scikit-learn *Pipeline* with feature standardisation *(StandardScaler)* applied inside each cross-validation fold to prevent data leakage. Cross-validated predictions were generated using *cross_val_predict* under the three CV schemes defined in Section 3. The best-performing model (XGBoost) was retrained on the full dataset and saved with *joblib* for deployment. Three additional XGBoost quantile regression models (q10, q50, q90) were trained to estimate predictive uncertainty and produce 80% prediction intervals.

4.2 Unsupervised Learning — Trial Network Optimisation

To evaluate redundancy among trial locations, a location-by-genotype yield matrix (5 × 237) was constructed from the aggregated dataset, with missing cells filled using column means. K-Means clustering was applied to the location profiles for k = 2, 3, and 4, with the optimal number of clusters selected by silhouette score. Hierarchical clustering with Ward linkage was applied as a second method to validate the cluster structure. A leave-one-location-out analysis was also performed using XGBoost: the model was iteratively retrained on four locations while holding one out, and the resulting 5-fold cross-validated RMSE was used to quantify each location's unique predictive contribution and identify potentially redundant trial sites.

4.3 Evaluation Metrics

Model performance was evaluated using three complementary metrics: Root Mean Square Error (RMSE, kg/ha) for prediction error in original units; coefficient of determination (R^2) for proportion of yield variance explained; and Pearson correlation coefficient (r) for linear agreement between observed and predicted values.

5. Results

5.1 Overall Model Performance

RMSE and R^2 for all six models across the three cross-validation schemes are presented in Table 1. AMMI-2 performed poorly across all schemes, with R^2 near zero under CV1 and negative under CV2. All five ML models substantially outperformed AMMI-2. XGBoost achieved the best performance overall with Random Forest close behind. Pearson correlation coefficients for XGBoost were r = 0.843 (CV1), r = 0.819 (CV2), and r = 0.847 (CV0).

Table 1. Model performance across three cross-validation schemes.

Model	CV1		CV2		CV0	
	RMSE	**R^2**	**RMSE**	**R^2**	**RMSE**	**R^2**
AMMI-2	1,773	0.013	1,886	-0.116	1,664	0.131
Ridge Regression	1,147	0.587	1,178	0.565	1,145	0.588
Lasso Regression	1,147	0.587	1,175	0.567	1,146	0.588
SVR (RBF kernel)	1,323	0.451	1,336	0.440	1,303	0.467
Random Forest	994	0.690	1,052	0.653	990	0.692
XGBoost	959	0.711	1,024	0.671	949	0.718

5.2 Observed vs Predicted Yield

Figure 1 presents observed versus predicted grain yield for the best-performing model (XGBoost) under all three CV schemes. Predictions closely follow the 1:1 reference line under CV1 and CV0, while slightly higher dispersion is observed under CV2.

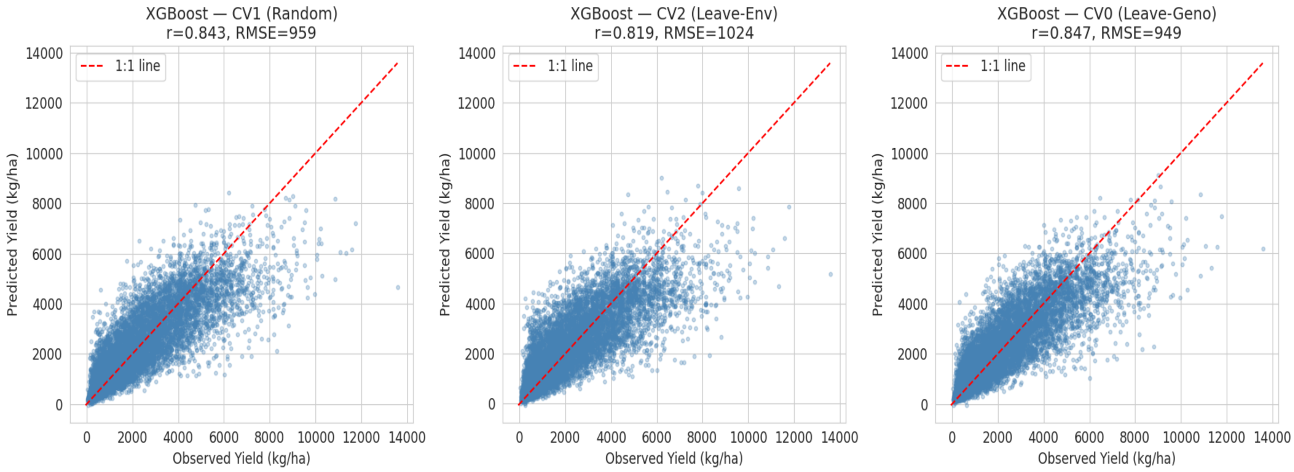

Figure 1. Observed vs Predicted Grain Yield for the XGBoost Model under Three Cross-Validation Schemes

5.3 Feature Importance Analysis

Feature importance analysis from Random Forest and XGBoost is shown in Figure 2. Both models consistently identified days to silking, stress tolerance index, ear aspect, and staygreen among the four most influential predictors of grain yield, though their relative ranking differed between models.

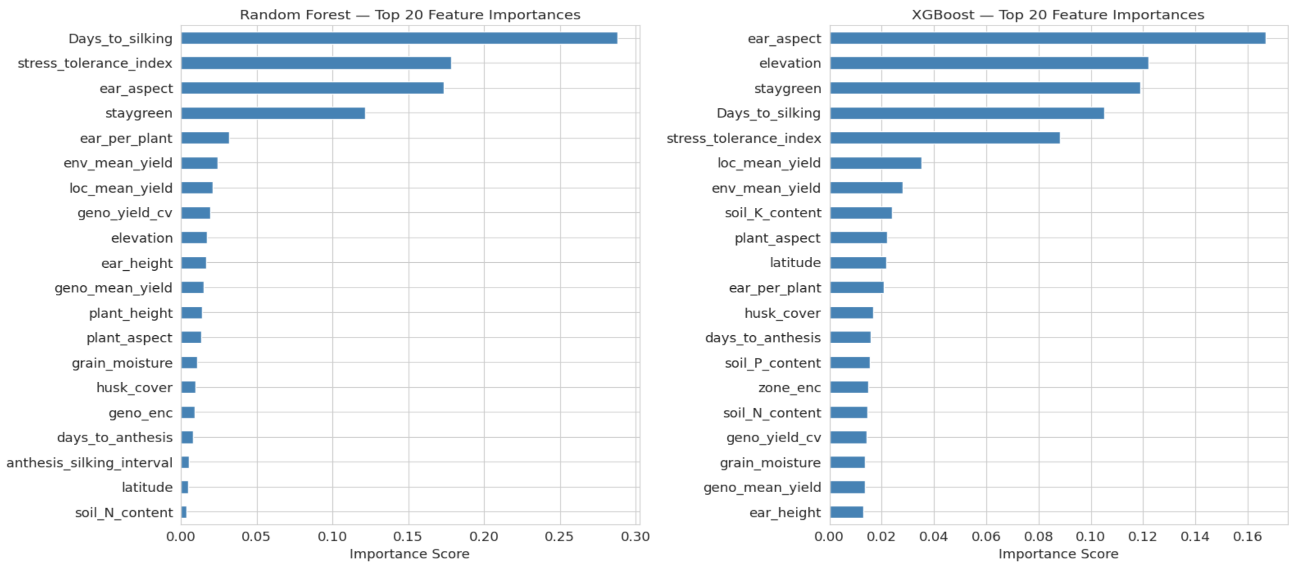

Figure 2. Feature Importance Ranking from Random Forest and XGBoost Models for Grain Yield Prediction

5.4 Location Clustering and Trial Network Optimisation

The drop-one-location analysis showed small changes in RMSE for most sites, with Omu-Aran decreasing slightly, Ibadan, Zaria, and Ilorin increasing by less than 1%, and Mokwa showing the largest increase (6.77%) (Table 4). Hierarchical clustering of trial locations based on genotype response patterns is shown in Figure 3. K-Means clustering identified an optimal value of k = 2 (silhouette score = 0.128), with Mokwa assigned to one cluster and Ibadan, Ilorin, Omu-Aran, and Zaria grouped in the second cluster.

Table 2. Drop-one-location analysis. Baseline RMSE (all five locations): 959.1 kg/ha.

Location Dropped	RMSE Without (kg/ha)	RMSE Change (kg/ha)	Change (%)
Omu-Aran	958.3	-0.8	-0.08%
Ibadan	962.4	+3.3	+0.34%
Zaria	963.2	+4.1	+0.43%
Ilorin	965.2	+6.1	+0.64%
Mokwa	1,024.1	+64.9	+6.77%

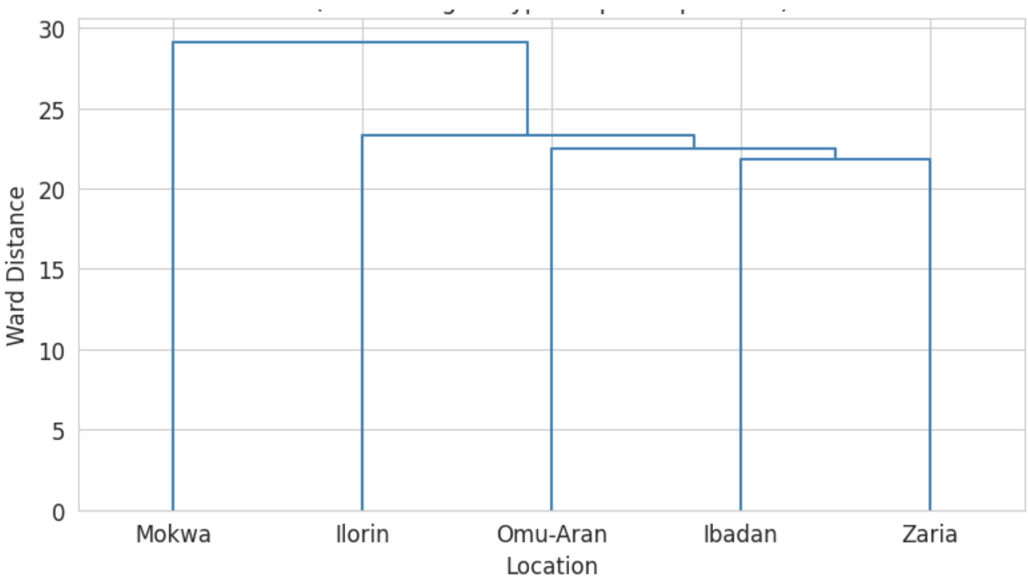

Figure 3. Hierarchical Clustering of Trial Locations Based on Genotype Response Patterns

6. Analysis

The results demonstrate that ensemble-based machine learning models are significantly more effective than classical statistical approaches for modelling G×E interactions in this dataset. The near-zero R^2 values for AMMI-2 under random cross-validation and the negative R^2 under leave-one-environment-out indicate that a rank-2 linear decomposition of the G×E matrix explains little of the yield variation. This suggests that the underlying G×E structure is largely non-linear and of higher complexity than can be captured by a low-rank AMMI formulation, which assumes predominantly linear multiplicative interactions.

XGBoost consistently outperformed all other models across validation schemes, reflecting its ability to capture non-linear relationships and higher-order interactions among genetic, environmental, and agronomic variables. This is consistent with findings reported in comparable crop yield prediction studies by Demir et al. (2022), Sahin (2023), and Ma et al. (2025). Engineered features such as genotype mean yield and stress tolerance index were among the most important predictors in both Random Forest and XGBoost models, highlighting the benefit of incorporating biologically informed aggregations.

The three CV schemes reveal a nuanced picture of prediction difficulty. The observed reduction in performance from CV1 to CV2 indicates that generalisation to unseen environments is more challenging than random splitting suggests, due to additional environmental heterogeneity. However, the relatively moderate increase in RMSE for XGBoost (959 → 1,024 kg/ha) suggests reasonable robustness to environmental shifts. In contrast, CV0 results indicate that prediction of unseen genotypes remains stable and is even slightly improved relative to random splits (959 → 949 kg/ha), suggesting that the models rely more on environmental and trait-based signals than on genotype memorisation. This implies that the modelling framework captures transferable agronomic and environmental relationships rather than overfitting to specific genotypes, which is desirable for real-world breeding applications where prediction on new varieties is required.

The location contribution analysis reveals partial redundancy among trial sites, with most locations providing overlapping environmental information, as reflected in minimal performance changes when individual sites are removed. The five-location network effectively separates into two distinct environmental groups: Mokwa and the remaining four locations. Removing Omu-Aran, Ibadan, Zaria, or Ilorin results in only minor increases in RMSE (less than 1%), whereas exclusion of Mokwa leads to a larger increase (6.77%). From a predictive standpoint, the network could potentially be reduced to two or three representative locations without substantial loss of accuracy, particularly by retaining Mokwa alongside one site from the larger cluster. This reduction could improve trial efficiency and resource allocation, enabling greater investment in genotype evaluation and expansion to additional environments.

7. Deployment

A prototype Streamlit web application was developed using the retrained XGBoost model as the primary prediction engine for grain yield estimation. The application provides tools for data exploration, model evaluation, yield prediction, and trial network optimisation. It is organised into six pages:

- Overview: project context, dataset summary statistics, and navigation guide.
- Upload Your Data: allows users to upload their own multilocational trial CSV, map column names to expected variables, and trigger retraining of the XGBoost model on their data. All subsequent pages then operate on the uploaded dataset.
- Data Explorer: interactive visualisation of trait distributions (violin plots), correlations with grain yield, and genotype performance comparisons across stress conditions.
- Model Validation: presents the pre-computed CV results (RMSE, R^2, Pearson r tables and charts) and XGBoost feature importance. For uploaded datasets, live CV evaluation is run.
- Yield Predictor: Supports three prediction scenarios: (1) known trial location, (2) new untested location (user supplies environmental covariates), and (3) new genotype (user supplies trait measurements). All predictions include an 80% quantile regression prediction interval.
- Location Clustering: applies K-Means and hierarchical clustering dynamically to the active dataset, displays results on an interactive folium map, silhouette analysis, dendrogram, and drop-one RMSE cost-benefit analysis.

The source code for the application is available in the project's GitHub repository. To run the application locally, the dataset CSV file should be placed in the same directory as the application files. Required dependencies can be installed using:

pip install streamlit pandas numpy scikit-learn xgboost matplotlib seaborn folium streamlit-folium plotly

The application can then be launched with:

streamlit run app.py

8. References

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9. Team Contributions

Team Member	Contributions
Olawale Serifdeen Aboderin (29206)	Dataset acquisition and curation; data cleaning and preprocessing pipeline; feature engineering; design and implementation of cross-validation schemes; model training, evaluation, and comparison; location clustering analysis; interpretation of results and agronomic discussion; report writing; Streamlit deployment application.
Francis Chinaecherem Uzor (29260)	Data cleaning and preparation; model training and evaluation; contribution to Streamlit deployment application; review and final proofreading of the report.

Appendix A. Overview of the maize trial dataset.

Attribute	Details	Count / Range
Years	2020, 2021, 2022	3 seasons
Locations	Ibadan, Ilorin, Mokwa, Omu-Aran, Zaria	5
Agroecological Zones	Northern Guinea Savanna, Southern Guinea Savanna, Forest–Savanna Transition Zone	3
Management Conditions	Optimum, Drought, Low-Nitrogen	3
Genotypes	Maize varieties evaluated across the trial network	237
Raw Observations	Individual plot records, 2 replications per genotype–environment combination	21,330
Observations After Aggregation	One row per genotype–environment combination	10,710
Missing Values	husk_cover (5), ear_harvested (5), ear_aspect (145), staygreen (1);	156 (0.019%)
Target Variable	Grain yield, adjusted to standard moisture, per-hectare equivalent	kg/ha

Appendix B: Dataset Variables and Descriptions

Variable	Description	Data Type	Unit
Agronomic Traits			
days_to_anthesis	Days from planting to anthesis	Numeric	Days
days_to_silking	Days from planting to silking	Numeric	Days
anthesis_silking_interval	Difference between anthesis and silking dates	Numeric	Days
plant_height	Plant height	Numeric	cm
ear_height	Height of ear placement	Numeric	cm
husk_cover	Husk cover rating	Ordinal	Score
plant_aspect	Overall plant appearance rating	Ordinal	Score
field_weight	Field weight of harvested ears	Numeric	kg
ear_harvested	Number of ears harvested	Numeric	Count
ears_per_plant	Number of ears per plant	Numeric	Count
ear_aspect	Ear appearance rating	Ordinal	Score
grain_moisture	Grain moisture content	Numeric	%
grain_yield	Grain yield (target variable)	Numeric	kg/ha
staygreen	Stay-green rating	Ordinal	Score
Environmental and Soil Variables			
latitude	Latitude of trial site	Numeric	Degrees
longitude	Longitude of trial site	Numeric	Degrees
elevation	Elevation of trial site	Numeric	m above sea level
rainfall_mm	Annual rainfall	Numeric	mm
mean_temperature	Mean annual temperature	Numeric	°C
soil_type	Soil type classification	Categorical	–
soil_pH	Soil pH	Numeric	pH
soil_N_content	Soil nitrogen content	Numeric	% or g/kg
soil_P_content	Soil phosphorus content	Numeric	mg/kg
soil_K_content	Soil potassium content	Numeric	cmol/kg
Management and Trial Variables			
YEAR	Trial year	Categorical	–
region	Trial location	Categorical	–
agro_ecological_zone	Agroecological zone of the trial site	Categorical	–
season	Trial season	Categorical	–
environment_condition	Trial management condition (optimum, drought, or low-nitrogen)	Categorical	–
rep	Replication number	Categorical	–
Genotype Information			
ENTRY	Entry number in the trial	Categorical	–
Pedigree	Genotype pedigree	Categorical	–
Name	Genotype or variety name	Categorical	–
breeding_institution	Source breeding institution	Categorical	–
maturity_group	Maturity group classification	Categorical	–
Engineered Features			
geno_mean_yield	Mean yield of a genotype across environments	Numeric	kg/ha
env_mean_yield	Mean yield of an environment across genotypes	Numeric	kg/ha
loc_mean_yield	Mean yield of a location across years and conditions	Numeric	kg/ha
stress_tolerance_index	Index quantifying performance under stress relative to optimum conditions	Numeric	Dimensionless
geno_yield_cv	Coefficient of variation of genotype yield across environments	Numeric	%
genotype_encoded	Encoded genotype identifier	Numeric	–
region_encoded	Encoded location identifier	Numeric	–
agro_zone_encoded	Encoded agroecological zone	Numeric	–
season_encoded	Encoded season	Numeric	–
environment_condition_encoded	Encoded management condition	Numeric	–
breeding_institution_encoded	Encoded breeding institution	Numeric	–
maturity_group_encoded	Encoded maturity group	Numeric	–

Appendix C: Hyperparameter Settings

Model	Hyperparameters
AMMI-2	n_components = 2
Ridge	alpha = 1.0
Lasso	alpha = 1.0, max_iter = 5000
SVR	kernel = RBF, C = 10, epsilon = 0.1
Random Forest	n_estimators = 200, max_depth = 10, min_samples_leaf = 5
XGBoost	n_estimators = 300, max_depth = 6, learning_rate = 0.05, subsample = 0.8, colsample_bytree = 0.8